Week 4
Lecture 6
Social Interaction

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Overview

- What is social interaction?
- The social mechanisms used in conversations
- What is meant by social presence?
- Overview of technologies for supporting social interactions
- How has social media changed the way we keep in touch?
- New social phenomenon arising from being able to connect online

Social interaction (SI)

- We live together, work together, play together, talk to each other, and socialize

- Social technologies developed to enable us to persist in being social when apart
  - They differ in how they support us
  - Some encourage social interactions (for example, family games with Alexa)
  - Others have a negative impact on everyday conversations (Turkle, 2015)…

SI: Are we spending too much time in our own digital bubbles?

SI: Questions raised by social tech

- Are F2F conversations being superseded by social media interactions?
- How many friends do you have on Facebook, LinkedIn, WhatsApp, and so on versus real life?
- How much do they overlap?
- How are the ways that we live and interact with one another changing?
- Are the established rules and etiquettes still applicable to online and offline?

SI: Conversational mechanisms

Various mechanisms and ‘rules’ are followed when holding a conversation face to face, such as mutual greetings

A: Hi there
B: Hi!
C: Hi
A: All right?
C: Good, how’s it going?
A: Fine, how are you?
C: OK
B: So-so. How’s life treating you?

SI: Conversational rules

Sacks et al. (1978) conversation analysis of conversations propose three basic rules:

Rule 1:

The current speaker chooses the next speaker by asking an opinion, question, or request

Rule 2:

Another person decides to start speaking

Rule 3:

The current speaker continues talking

SI: More conversational rules

Turn-taking used to coordinate conversation

A: Shall we meet at 8:00?
B: Um, can we meet a bit later?

A: Shall we meet at 8:00?
B:Wow, look at him?

A:Yes what a funny hairdo!
B: Um, can we meet a bit later?

Back channeling to signal to continue and following

- Uh-uh, umm, ahh

SI: Further conversational rules

Farewell rituals

- Bye then, see you, yeah bye, see you later….

Implicit and explicit cues

- For instance, looking at watch or fidgeting with coat and bags
- Explicitly saying, "Oh dear, look at the time, I must go, I'm running late…"

When someone says something that is misunderstood:

- Speaker will repeat with emphasis:

  A: "This one?"
  B: "No, I meant that one!"

SI: What happens in online conversations?

- **Do the same conversational rules apply?**
- **Are there different kinds of breakdowns?**
- How do people repair them for:
  - Email?
  - Instant messaging?
  - Texting?
  - Skype or other videoconferencing software?

Remote conversations (RC)

- Much research on how to support conversations when people are ‘at a distance’ from each other
- Many applications have been developed
  - For example, email, videoconferencing, instant messaging, and chatrooms
- Do they mimic or move beyond existing ways of conversing?

RC: VideoWindow system (Bellcore, 1989)

- Shared space that allowed people 50 miles apart to carry on a conversation as if in same room drinking coffee together
- 3 x 8 foot ‘picture-window’ between two sites with video and audio
- People did interact via the window, but strange things happened (Kraut, 1990)

RC: Diagram of VideoWindow in use

RC: Findings of how VideoWindow System was used

- Talked constantly about the system
- Spoke more to other people in the same room rather than in other room
- When trying to get closer to someone in the other place it had opposite effect—participants went out of range of the camera and microphone
- No way of monitoring this

RC: Video conferencing and telepresence rooms

- Many to choose to connect multiple people (for instance, Zoom)

- Customized telepresence rooms for groups

RC: Telepresence robots

Enable people to attend events who could not do so, such as by controlling their robot remotely

- In places such as schools, conferences, and museums
- Early example: Beam+
- Often dressed up to appear like the person to others at the event
- Positive experience of being there

Susan Lechelt at ACM CHI

RC: Social Presence - Facebook's vision of socializing in a 3D world using VR

- Two avatars talking at a virtual table
- Users experience each other through donning VR headsets

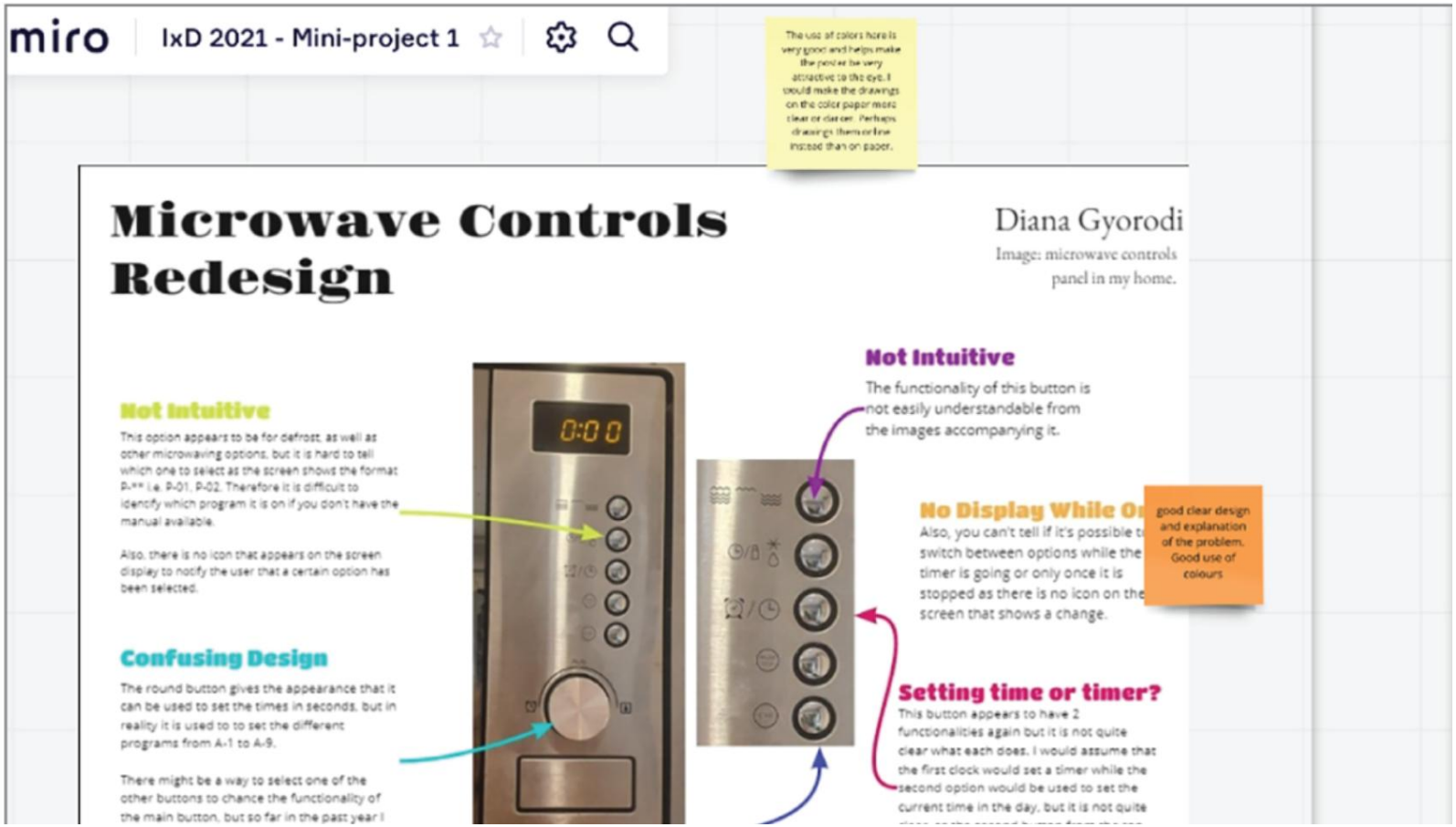
RC: Social Presence - Facebook's vision of socializing in a 3D world usingVR

- How much realism and immersion are necessary..?

RC: Social Presence – Second Life Example

RC: Collaborative Tools – MIRO Example

RC: Collaborative Tools – MIRO Example (2)



RC: Co-presence

- Co-located groups/people who want to collaborate
- Many technologies have been designed to:
  - Enable groups to work, learn and socialize more effectively together
  - For example, tabletops, whiteboards, and people-in-the-box

RC: Co-presence – Physical Coordination

- When a group of people act or interact together, they need to coordinate themselves
  - For example, when playing football or navigating a ship
- To do so, they use:
  - Verbal and non-verbal communication
  - Schedules, rules, and conventions
  - Shared external representations

RC: Co-presence - Awareness mechanisms

- Involves knowing who is around, what is happening, and who is talking with whom (Dourish and Bly, 1992)
- Peripheral awareness
  - Keeping an eye on things happening in the periphery of vision
  - Overhearing and overseeing—allows tracking of what others are doing without explicit cues
- Situational awareness
  - Being aware of what is happening around you in order to understand how information and your actions will affect ongoing and future events
  - For example, air traffic control or an operating theatre

RC: Co-presence - The Reflect Table

- LEDs lit up to reflect how much each member of the group spoke
- Used microphones in front of each individual to do this
- Study showed those who spoke the most changed their behavior the most
- Those who spoke the least did not change their behavior
- Why do you think this is?

RC: Co-presence - Sococo floor plan of a virtual office: who is where and who meeting with whom

See a team in a meeting, sharing screens and viewing documents in a room

Name a room to reflect the topic of a meeting in progress

Search for colleagues across the workspace to see status or instantly chat

Instantly "Get" colleagues to spontaneously collaborate

Scale instantly with unlimited floors to your Sococo space

Knock on a door to join a meeting or just pop in

Send a link for a guest to join you in your Sococo office

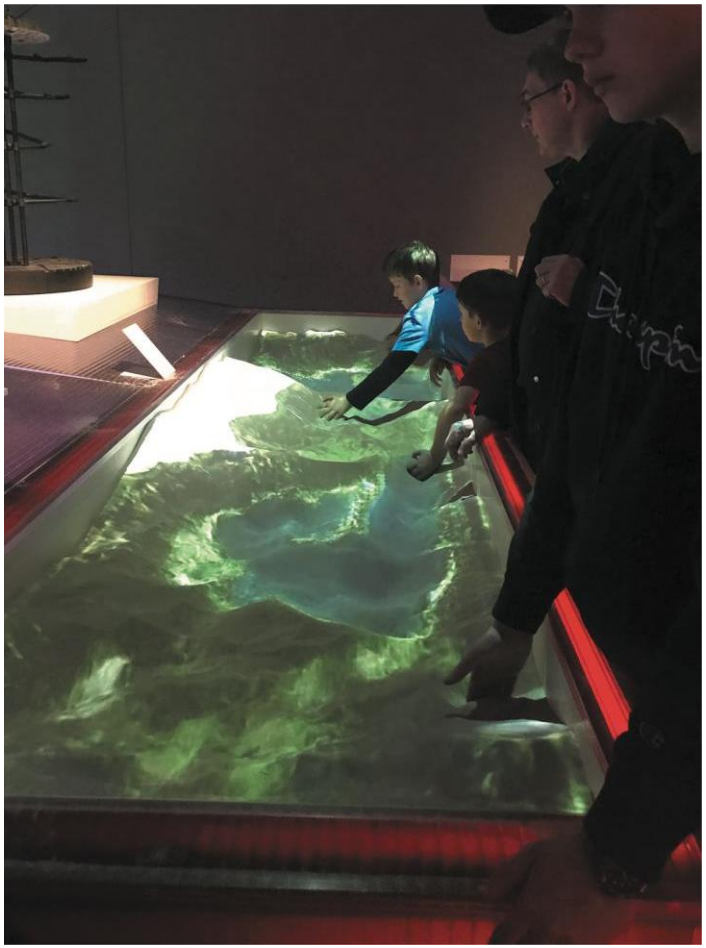
Blinking avatars are colleagues collaborating

Share documents or links on a desk for immediate access to anyone in the room

https://www.sococo.com/

RC: Co-presence - Playing together in same space

- Visitors using an AR sandbox at the V and A
- Visitors sculpt landscapes out of sand
- System reacts with changing superimposed digital colored landscape
- Enables creative forms of collaboration



Social engagement

- Refers to participation in activities of a social group
- Social exchange where people give or receive something from others
- Voluntary, unpaid and often altruistic (in the sense of sharing and doing good for others)
- Websites often used as hub to connect people
- Retweeting is a powerful way of connecting millions of people…

Social Engagement: Dilemma: Is it OK to talk with a dead person using a chatbot?

- Eugenia Kuyda lost a close friend in a car accident who was only in his 20s
- She took all his texts sent over the course of his life and made a chatbot using them
- Chatbot responds to text messages so that Eugenia can talk to her friend as if he was alive
- Is this a creepy or comforting way to deal with grief?
  - Is it respectful of the dead person?

Summary

- Social interaction is central to our everyday lives
- Social mechanisms, like turn-taking, enable us to collaborate and coordinate our activities
- Keeping aware of what others are doing and letting others know what you are doing are important aspects of collaborative working and socializing
- Many technology systems have been built to support telepresence, social presence, and co-presence
- Social media has brought about significant changes in how people keep in touch and manage their social lives